

Frame It as Cases

Voice Card

Direct, honest, practical, simple, no buzzwords.

Content Refresh Opportunity Scoring (In Progress)

Project Status

In Progress

The Problem

Content teams can't manually review every page to decide what needs updating because it doesn't scale. The goal of this project is to use search and engagement data to build a ranked priority list so the content team can spend their time reviewing the pages that need attention the most instead of checking every page individually.

What I Had to Work With

The starter dataset included metrics such as impressions, clicks, CTR, sessions, engagement rate, trend direction, search volume, content age, days since last update, average position, and content type.

I decided to focus on six signals first: trend direction, CTR, engagement rate, days since last update, impressions, and clicks. I chose these because they seemed most closely related to how a page is currently performing and whether it might need to be reviewed. I did not ignore the other features, but I wanted to begin with the signals that were directly connected to page performance before exploring additional factors.

What I Did

I explored the dataset using Python and pandas. I checked the dataset size, column names, sample records, and summary statistics to understand the available information. I also tested one of my assumptions by checking whether search volume and impressions were closely related. The correlation was almost zero, which showed that search volume alone was not a reliable indicator of impressions in the starter dataset.

I completed two notebooks during this stage of the internship.

The first notebook focused on defining the research question, selecting my project lane, identifying the unit of analysis, and supporting those decisions with exploratory data analysis.

The second notebook framed the project as a machine learning task. I defined the target as a content refresh priority score that represents how urgently a page should be reviewed. I also explained why this problem could benefit from machine learning while still planning to build a simple rule-based baseline before comparing different approaches.

Decisions I Made

I decided to understand the problem before selecting any machine learning model. Instead of searching for the best algorithm immediately, I first focused on defining the research question, identifying the target, understanding the data, and selecting the signals that were most useful for this stage of the project. I also chose to build a simple baseline first so I would have something meaningful to compare with future machine learning models.

What Was Hard

The biggest challenge was realizing that choosing an algorithm was not the first step. At the beginning I was more focused on machine learning models than on understanding the actual decision the project needed to support. I also found it difficult to separate performance signals from contextual information such as search volume and content age. Working through the assignments helped me understand that these decisions should be based on evidence from the data rather than assumptions.

What I Learned

One of the biggest lessons from this project is that a single metric cannot explain page performance. I originally thought that metrics like search volume or impressions would be enough to identify pages that needed a refresh. After exploring the data, I realized that different signals need to be considered together because each one provides only part of the picture. This changed the way I think about building a content refresh scoring system.

What's Next

The next step is to build a simple rule-based baseline using the six selected signals and evaluate whether the ranking is useful for prioritizing pages. After that, I plan to compare the baseline with machine learning approaches to see whether they produce better recommendations. I also want to develop a more objective way to evaluate the quality of the ranking instead of relying only on manual observation.

About Me

I am a Computer Science student with an interest in Machine Learning and Artificial Intelligence. Through the FlyRank ML Internship, I am learning how to frame machine learning problems, work with real datasets, and build practical solutions one step at a time. My goal is to become an AI and Machine Learning Engineer by strengthening both my technical skills and problem-solving approach through real-world projects.

Contact

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Before and After

Before

I leveraged machine learning techniques to build an innovative content optimization solution that improves business outcomes through data-driven insights.

After

I explored search and engagement data to understand which pages might need a content refresh. Before building any machine learning model, I focused on understanding the problem, exploring the data, selecting useful signals, and creating a baseline that I can compare with future models.